Ian C. George

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Waterloo, ON
Canada

Education

2023- PhD Combinatorics and Optimization, University of Waterloo. Supervisor: Rowan Yeats.

2021-2023 MSc Mathematics, Dalhousie University. Supervisor: Jason Brown.
2017-2021 BSc Mathematics (Honours) and Physics with certificate in Information Technology, Dalhousie University.

Publications

Journal articles

2025 Jason I. Brown and Ian C. George. On the roots of degree polynomials. *Australasian Journal of Combinatorics*, 92(2):164–183, 2025.
2024 Jason I. Brown and Ian C. George. Modulation Graphs in Popular Music. *The Mathematical Intelligencer*, pages 1–8, 2024.

Preprints

2026 Ian George and Rowan Yeats. A Generating Polynomial for Convex Subsets of Posets. *arXiv:2610.23399*, 2026.
2026 Matteus Froese and Ian George. Path Covering Spaces of Posets. *arXiv:2610.20890*, 2026.
2025 Ian George and Karen Yeats. The Generalized Interval Polynomial of a Poset. *arXiv:2511.14924*, 2025 (extended abstract submitted to FPSAC 2026).

Theses

2023 Ian George. On the Degree Polynomial of Graphs. Master's thesis, Dalhousie University, 2023.

Teaching and related experience

Mentoring/supervision

2026 Mentor for the Women in Math (WiM) Directed Reading Program (DRP), University of Waterloo. Supervised undergraduate students on a research project investigating interval partitions of posets.

2025 Mentor for the Women in Math (WiM) Directed Reading Program (DRP), University of Waterloo. Supervised undergraduate students on a research project investigating poset invariants in Causal Set Theory (CST).

TA experience

2023- TA for undergraduate mathematics courses, University of Waterloo. Courses: MATH 239, 136, 225, CO 480, 330. Duties involved grading student assignments, tests, and delivering tutorials.

2021-2023 Learning centre TA, Dalhousie University. Helping undergraduate students with course work in various undergraduate courses.

2021-2023 TA for undergraduate mathematics courses, Dalhousie University. Courses: MATH 2110, 1280, 1010. Duties involved grading assignments, tests, and delivering tutorials.

Grants, honours & awards

2024 Combinatorics and Optimization Graduate Award (University of Waterloo)

2023-2024 Math Domestic Graduate Student Award (University of Waterloo)

2022-2023 NSERC Canada Graduate Scholarships - Master's (CGS-M)

2021 NSERC Undergraduate Summer Research Award (USRA)

2020 Faye Sobey Undergraduate Research Award, Sobey Foundation

2019 Archibald Award, Department of Physics and Atmospheric Science, Dalhousie University

2019 A.S. MacKenzie Award, Department of Physics and Atmospheric Science, Dalhousie University

2019 Faye Sobey Undergraduate Research Award, Sobey Foundation

2018 J.G. MacGregor Award, Department of Physics and Atmospheric Science, Dalhousie University

2017 Entrance scholarship, Dalhousie University

Talks and Posters

Talks

2026 *Some Poset Invariants for Causal Sets*, in the Theoretical Physics seminar, DIAS, Dublin.

2026 *Enumerating Convex Sets of Posets*, in the Algebraic and Enumerative Combinatorics Seminar, University of Waterloo.

Posters

2026 *Enumerating Convex Sets in Posets*, Algebraic Combinatorics Virtual Expedition (AlCoVE) online conference.

Outreach/volunteering

2024 Co-organizer for MITP YOUNGST@RS *Combinatorics in Fundamental Physics* virtual workshop. Located virtually at the Mainz Institute for Theoretical Physics, Johannes Gutenberg University,

November 26-29. Held invited talks about the role of combinatorics in several areas of fundamental physics, and had a cross-disciplinary discussion session.